

ECO 362 Financial Investments

Precept 7: Arbitrage Pricing Theory and Alpha

November 3 – 6, 2025

- APT: Multi-factor model of stock returns
- Efficient market hypothesis
- Alpha: Performance evaluation

1 Arbitrage Pricing Theory (APT)

2 Efficient Market Hypothesis

3 Alpha

- A general multi-factor model:

$$\mathbb{E}[r_i - r_b] = \sum_j \beta_{ij} \mathbb{E}[F_j], \quad (1)$$

where

- β_{ij} is the quantity of risk factor j
 - $\mathbb{E}[F_j]$ is the price of risk factor j
- CAPM as a one-factor model:

$$\mathbb{E}[r_i - r_b] = \beta_{i,m} \mathbb{E}[r_m - r_b]. \quad (2)$$

- 3 factors:
 - 1 Market factor: Market minus riskless
 - 2 Size factor: Small minus Big (SMB) market equity
 - 3 Value factor: High minus Low (HML) book-to-market equity
- Fama-French 3-factor model:

$$\mathbb{E}[r_i - r_b] = \beta_{i,m}\mathbb{E}[r_m - r_b] + \beta_{i,SMB}\mathbb{E}[SMB] + \beta_{i,HML}\mathbb{E}[HML], \quad (3)$$

- Testing the model (regression):

$$r_{it} - r_{bt} = \alpha_i + \beta_{i,m}(r_{mt} - r_{bt}) + \beta_{i,SMB}SMB_t + \beta_{i,HML}HML_t + \varepsilon_{it} \quad (4)$$

Hypothesis: $H_0 : \alpha_i = 0$ vs $H_1 : \alpha_i \neq 0$.

Interpreting Fama-French 3-factor model:

$$r_{it} - r_{bt} = 0.03\% + 1.98(r_{mt} - r_{bt}) + 0.21\text{SMB}_t - 0.04\text{HML}_t + \varepsilon_{it} \quad (5)$$

Portfolio strategy (\$1 wealth):

- $\beta_m = 1.98$: Long \$1.98 in market and short \$0.98 in riskless asset.
- $\beta_{SMB} = 0.21$: Long \$0.21 of small-cap stocks and short \$0.21 of large-cap stocks.
- $\beta_{HML} = -0.04$: Short \$0.04 of value stocks (high book-to-market) and long \$0.04 of growth stocks (low book-to-market).
- Overall: small-cap growth strategy.

Excel Examples 2 and 3

Value and momentum anomalies

1 Arbitrage Pricing Theory (APT)

2 **Efficient Market Hypothesis**

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Forms of market efficiency:

- Weak form: Asset prices reflect information that can be inferred from historical prices and returns.
- Semi-strong form: Asset prices reflect all publicly available information.
- Strong form: Asset prices reflect even private or “insider” information.

Event study procedures:

- Estimate CAPM on daily returns prior to the event date (assuming constant riskless rate):

$$r_{it} = \alpha_i + \beta_{i,m} r_{mt} + \varepsilon_{it} \quad (6)$$

- Use estimates $\hat{\alpha}_i$ and $\hat{\beta}_{i,m}$ to compute abnormal returns for the whole sample:

$$AR_{it} = r_{it} - (\hat{\alpha}_i + \hat{\beta}_{i,m} r_{mt}) \quad (7)$$

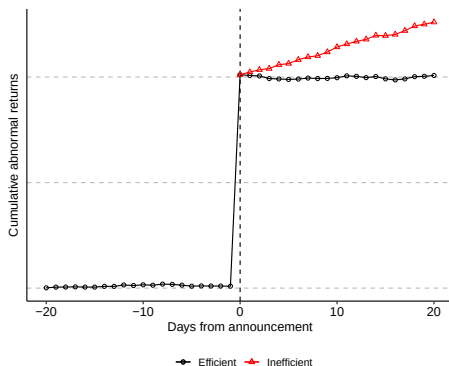
Abnormal returns isolate the impact of the event (idiosyncratic) from market fluctuations (systematic).

- Compute/Plot cumulative abnormal returns around event date.

Efficient-Market Hypothesis: Examples

Positive Earnings Announcement:

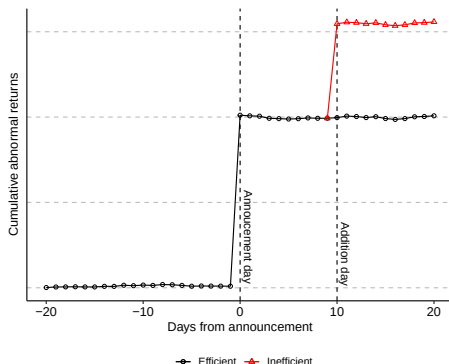
- If market is efficient (semi-strong), cumulative abnormal return jumps up on the announcement date (to reflect public information), and stays flat afterwards.
- If abnormal return further drifts away after the announcement date, it may suggest market inefficiency.



Efficient-Market Hypothesis: Examples

Index addition:

- If market is efficient (semi-strong), cumulative abnormal return jumps up on the announcement date (to reflect public information), and stays flat afterwards.
- If abnormal return jumps up again on the addition day, it may suggest market inefficiency.



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- Benchmark selection:
 - Explicit: index fund vs the index
 - Peers: PRINCO vs Cambridge Associates
 - When not obvious, use Fama-French 3-factor model

- Regression:

$$r_{it} - r_{bt} = \alpha_i + \beta_i(\text{Benchmark}_t - r_{bt}) + \epsilon_{it} \quad (8)$$

- α_i : average performance relative to the benchmark
 - β_i : risk relative to the benchmark
 - ϵ_{it} : tracking error (deviations from the benchmark)
- Information ratio: $\frac{\alpha_i}{\sigma(\epsilon_{it})}$
 - Higher information ratio: higher relative return and lower tracking error